

QR Code Version	Module Size	ECC Level	Alphanumeric capacity (chars)	Total available payload, excluding prefix (bits)	Available payload for TLV data (bits)
1	21x21	L	25	104	16
2	25x25	L	47	208	120
		M	38	168	80
		Q	29	120	32
3	29x29	L	77	352	264
		M	61	272	184
		Q	47	208	120
		H	35	152	64
4	33x33	L	114	528	440
		M	90	416	328
		Q	67	304	216
		H	50	224	136
5	37x37	L	154	720	632
		M	122	568	480
		Q	87	400	312
		H	64	288	200

NOTE

Version 1 codes with ECC levels M, Q, and H and version 2 codes with ECC level H have insufficient capacity

NOTE

*Total available payload, excluding prefix = $(\text{trunc}((N-3) / 5) * 24)$ where **N** is the number of alphanumeric characters which fit in the QR code. This formula uses **N-3** to account for the prefix characters, and then determines how many groups of 5 base-38-encoded characters can fit; each such group carrying 24 bits of payload.*

*This formula fills groups of 5 characters after the **MT:** prefix. If there are 2,3 or 4 characters left after these groups, an additional 8 bits (for 2,3 characters) or 16 bits (for 4 characters) of TLV data can be accommodated. So the entries in the table take this into account.*

Available payload for TLV data = $(\text{Total available payload, excluding prefix} - 88)$ since the minimum payload for the Packed Binary Data Structure is 84 bits before padding, or 88 bits with padding.

Payload Limits

The number of alphanumeric characters in the QR code to represent the Onboarding Payload for a single product SHALL NOT exceed 255 characters. Using this maximum size would yield a total available payload of 1208 bits, and hence 1120 bits (140 octets) of TLV payload.